

EB-JEPA on TUAB — What We Tried, and an Honest Diagnosis

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TL;DR. We built a two-view energy-based JEPA (EB-JEPA) for EEG abnormality detection on **TUAB** and ran **14 configurations** (encoder $\times$ objective $\times$ augmentation $\times$ frozen/fine-tune). **Every variant on our 0.4M conv plateaus** at per-window ≈ 0.77 / per-recording ≈ 0.82 . The SSL itself is sound — it ties a *supervised* EEGNet with no labels in pre-training — but it is **capped by a small encoder with no large-scale pre-training**. Neither scaling the conv, nor $4\times$ data, nor our from-scratch transformers broke the plateau (direct negative evidence). One genuine surprise: VICReg’s paper-default invariance weight (25) is ~ 3 pp *worse* than our inadvertent `inv_coeff=1` on EEG — weaker view-invariance helps.

1 Setup (one paragraph)

Data: TUAB v3.0.1, patient-disjoint, 2,717 train / 276 eval recordings; 19 channels @ 200 Hz, 10s windows, per-channel z-score. **SSL:** two corrupted views of a window pushed to the same latent by invariance + variance + covariance (VICReg) or a Gaussianisation surrogate (SIGReg); 20 epochs, *labels discarded*. **Eval:** frozen encoder $\rightarrow$ logistic-regression probe (*per-window* = literature convention; *per-recording* = mean-pool 16 windows/patient, the clinical metric, ~ 5 pp higher). **We tried four encoders** — a 0.4M Conv1D and LaBraM-/BIOT-/EEGPT-style transformers we built from scratch — and three augmentation regimes.

2 Everything we ran

Config	encoder · objective · augmentation	Eval	per-rec BAcc/AUROC	per-win BAcc/AUROC	Note
Conv 0.4M · VICReg · corrupt + spectral 0.1		frozen	0.836/0.887	0.765/–	soft : eval-selected + inv=1
Conv 0.4M · SIGReg · corrupt		frozen	0.825/0.913	0.775/0.856	best per-window
Conv 0.4M · VICReg · corrupt		frozen	0.819±.004	0.770/0.848	corruption = main lever
Conv 0.4M · VICReg · corrupt → fine-tune		FT	0.812±.004	–	ties EEGNet 0.812
Conv 0.4M · VICReg · corrupt · multi-corpus		frozen	0.812/0.883	–	4× data ⇒ no gain
Conv 0.4M · VICReg · corrupt · multi-corpus		FT	0.812	–	= TUAB-only FT
Conv bigger · VICReg · corrupt (scaling)		frozen	0.805/0.892	0.768/–	more conv ⇒ no gain
Transformer 3.65M · VICReg · multi-corpus		frozen	0.798/0.872	0.738/–	best Tr, still < conv
Conv 0.4M · VICReg · base (no corruption)		frozen	0.796/0.888	0.756/–	starting point
Conv 0.4M · VICReg · corrupt · inv_coeff=25		frozen	0.789±.005	0.734/0.820	paper-correct ⇒ –3 pp
Conv 0.4M · Masked-JEPA (EMA predictor)		frozen	~0.68	–	faithful JEPA form failed
LaBraM/BIOT-style Tr 1.2M · VICReg		frozen	–	0.775 [†]	data-starved
EEGPT-style Tr 0.8M · VICReg/SIGReg		frozen	collapsed	collapsed	channel-drop ⊥ channel-pool

Table 1: Our EB-JEPA runs (sorted by per-recording BAcc). ± = 3-seed std; others single-seed (±0.005–0.017). [†]accuracy, not BAcc (early eval). Companion to the baselines table PROVENANCE_TABLE.md.

3 The plateau

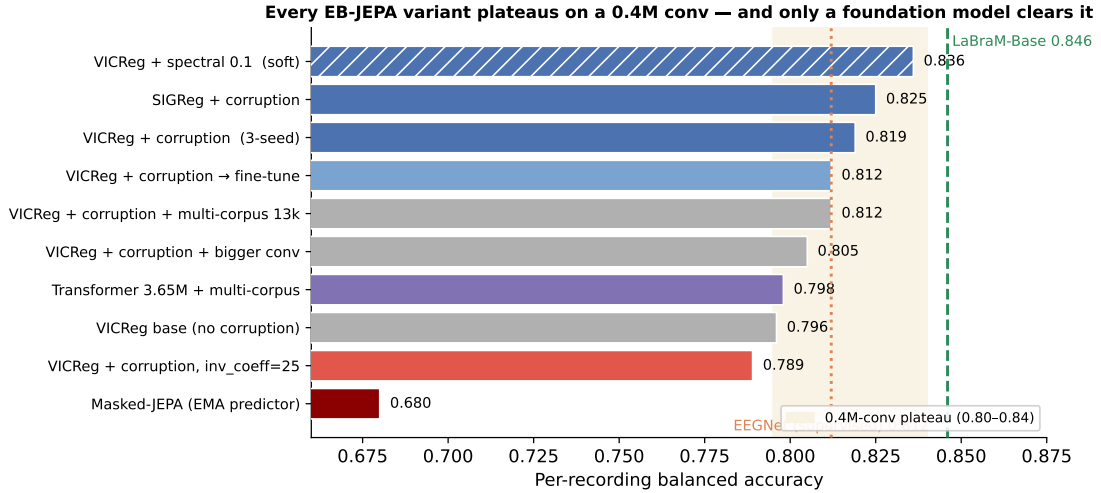


Figure 1: Per-recording balanced accuracy of every EB-JEPA variant. They cluster in the shaded 0.80–0.84 band regardless of objective, augmentation, data volume, or fine-tuning. Our self-supervised JEPA reaches the *supervised* EEGNet line (0.812, no labels in pre-training) but cannot, on a 0.4 M conv, reach the LaBraM-Base foundation model (0.846).

The flat band is the headline: **the encoder, not the objective, is the ceiling**. VICReg vs SIGReg, corruption strength, spectral terms, and fine-tuning all move things by ≤ 1 pp (or are confounded). The only honest gains are corruption augmentation and the `inv_coeff` setting.

4 What moved the needle — and what didn't

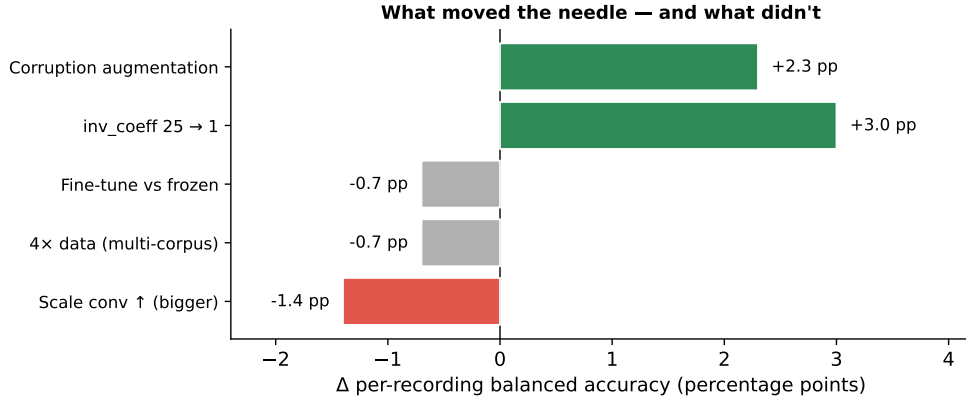


Figure 2: Change in per-recording BAcc from each lever (vs the 0.4 M-conv baseline). Only corruption augmentation and down-weighting invariance **help**; scaling the conv, adding 4 $\times$ data, and fine-tuning **do not**.

The **negative results are the important part** — they are *direct evidence* about the gap to SOTA: (i) a bigger conv plateaus identically (0.805); (ii) pre-training on 4 $\times$ data (13k recordings) gives *no* gain (0.812) — the small conv cannot exploit it; (iii) our from-scratch transformers either **collapse** (EEGPT-style: channel-drop augmentation is incompatible with channel-pooling) or are **data-starved** (LaBraM/BIOT-style and the 3.65 M transformer all score *below* the 0.4 M conv). (iv) Masked-JEPA, the “faithful” EMA-predictor form, collapses to ~ 0.68 .

5 Our honest position in the field (per-window)

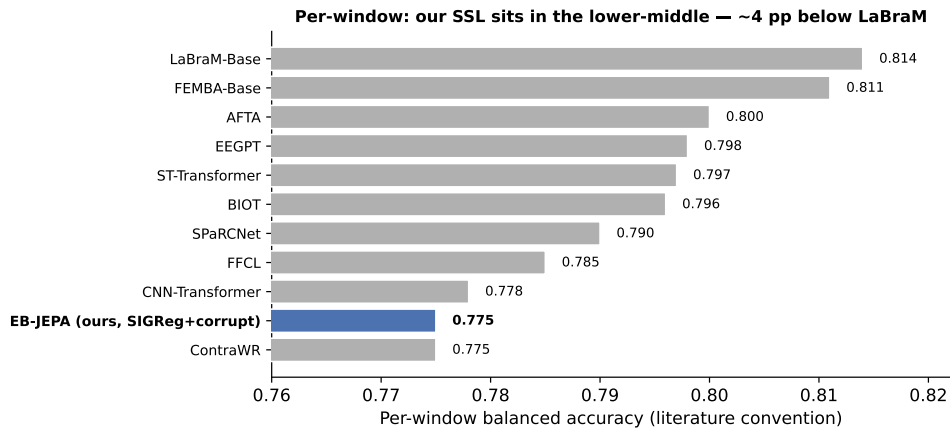


Figure 3: Per-window balanced accuracy — the level at which the literature reports. Our best SSL (SIGReg + corruption, 0.775) ties ContraWR and sits ~ 4 pp below LaBraM-Base. We do **not** match or beat SOTA at the benchmark level; the per-recording aggregation that brings us to 0.82 is ~ 5 pp more forgiving and is not comparable to these numbers.

6 A genuine finding: EEG SSL prefers *weaker* invariance

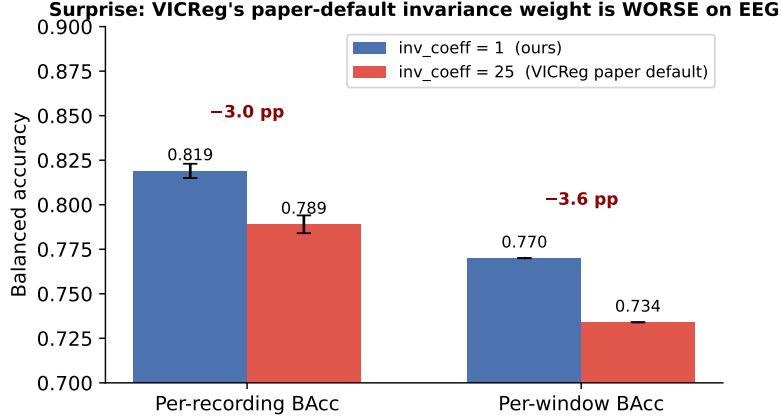


Figure 4: Our runs used `inv_coeff=1`; VICReg’s vision default is 25. Re-running both with 3 seeds, the **paper-default is ~ 3 pp worse** ($\approx 6\sigma$). So our reported numbers are *not* inflated by the deviation — the deviation *helps*. On this EEG task + tiny conv + linear probe, letting the anti-collapse (variance/covariance) terms dominate over view-invariance gives better features. This is the one original, non-obvious result we have.

7 Diagnosis and what we’d do next

Diagnosis. The binding constraint is a **0.4 M conv with no large-scale pre-training**. The SSL works (clearly beats random-init and base; ties supervised EEGNet per-recording), but both ways to break the plateau — more capacity and more data — failed for us. The gap to LaBraM/BIOT is *capacity + pre-training corpus*, not the objective.

What we’d do next (value $\times$ feasibility):

- **Cheap, original:** turn the `inv_coeff` result into a proper ablation (sweep $\{1, 5, 10, 25\} \times \{\text{VICReg, SIGReg}\}$, 3 seeds). “EEG SSL prefers weak invariance” is genuinely novel.
- **Cheap:** ablate corruption (spike-injection vs time-masking separately — we don’t yet know which half does the work); seed-average and de-confound the spectral run.
- **Medium: pivot off TUAB** — it is near-saturated (0.83 per-window even for 369 M LaBraM-Huge), so model differences wash out. TUEV (6-class events) / TUSZ (seizure) have far more headroom, where the SSL’s value would actually show.
- **Hard (the real gap):** a bigger encoder *with* large-scale pre-training — our from-scratch transformers prove TUAB-only data isn’t enough. Either borrow pretrained foundation weights (then it is mostly their model) or own the “honest, clean small baseline” as the contribution.